

Resistive circuit

Resistors

- 1- Dissipate energy
- 2- Applications limited

Passive linear circuits

capacitors | inductors

- 1- Store energy
(Storage elements)
- 2- More practical circuits.

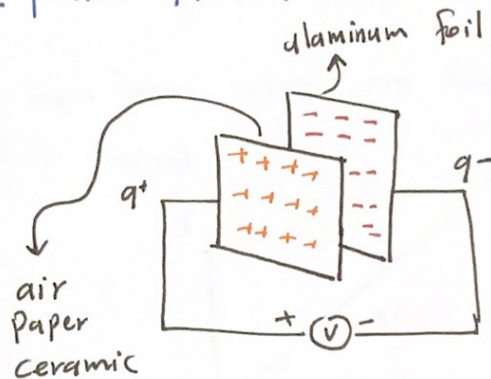
Capacitors $\Rightarrow$

Passive element, store energy in its electric field.
It consists of two conducting plates ~~per~~ separated by insulator (dielectric).

When connected:

The amount of charge stored q is proportional to V

$$q = CV$$



C : Capacitance, Farad = $\frac{\text{Coulomb}}{\text{Volt}}$

Also,

$$C = \frac{\epsilon A}{d}$$

Permittivity of dielectric ϵ $\rightarrow$ surface area of each plate A
 d $\rightarrow$ distance between plates

- 1- Large $A \Rightarrow$ greater capacitance
- 2- smaller spacing $d \Rightarrow$ greater capacitance
- 3- Higher permittivity $\epsilon \Rightarrow$ greater capacitance.

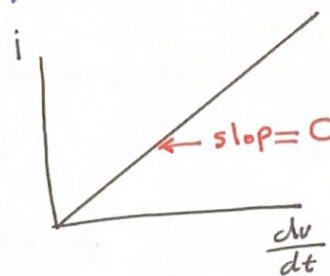
⇒ Capacitors are used to

1. Block dc
2. Pass AC
3. Shift phase
4. Store energy
5. Start Motor
6. Suppress Noise.

$$\begin{aligned} q &= C V \quad \text{differentiate} \Rightarrow \frac{dq}{dt} = C \frac{dv}{dt} \\ \text{We know } i &= \frac{dq}{dt} \\ i &= C \frac{dv}{dt} \end{aligned}$$

0- Current - Voltage Relationship

⇒ Although some capacitors are non linear, most are linear (only linear capacitor assumed in this course)



⇒ Another formula for the current-voltage relation by integrating

$$i = C \frac{dv}{dt} \quad \text{integrate} \quad V(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$$

$$V(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + V(t_0)$$

$V(t_0) = \frac{q(t_0)}{C}$: Voltage across capacitor at time t_0

$$\Leftrightarrow \boxed{V(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + V_{t_0}} : \text{The capacitor voltage}$$

depends on the past history of the capacitor current.

The capacitor has memory

$\Leftrightarrow$ Power delivered to the capacitor

$$P = Vi = V \cdot C \frac{dV}{dt}$$

$\Leftrightarrow$ Energy stored in the capacitor

$$W = \int_{-\infty}^t P(\tau) d\tau = C \int_{-\infty}^t V \frac{dV}{d\tau} d\tau = \int_{V(-\infty)}^{V(t)} V dV = \frac{1}{2} C V^2 \Big|_{V(-\infty)}^{V_t}$$

$V_{(-\infty)} = 0$ as uncharged capacitor at $t = -\infty$

$$\boxed{W = \frac{1}{2} C V^2 \quad \text{or} \quad W = \frac{q^2}{2C}}$$

Ex- 6.1, 6.2, P. 6.3, P. 6.4, 6.5

Series and Parallel Capacitors

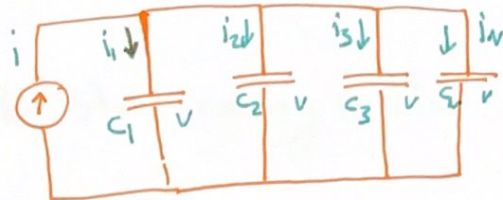
A- Parallel-connection

$$i = C \frac{dv}{dt}$$

$$i = i_1 + i_2 + i_3 + \dots + i_N \quad (\text{KCL})$$

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \dots + C_N \frac{dv}{dt}$$

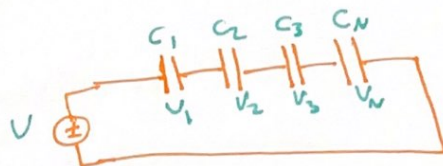
$$= \underbrace{(C_1 + C_2 + \dots + C_N)}_{C_{eq}} \frac{dv}{dt}$$



The equivalent capacitance of N parallel-capacitors is the sum of them.

B- Series-connection

$$V = V_1 + V_2 + V_3 + \dots + V_N$$



For each capacitor $V = \frac{1}{C} \int_{t_0}^t i \tau d\tau + V_{(t_0)}$

$$V = \frac{1}{C_1} \int_{t_0}^t i \tau d\tau + V_1(t_0) + \frac{1}{C_2} \int_{t_0}^t i \tau d\tau + V_2(t_0) + \dots + \frac{1}{C_N} \int_{t_0}^t i \tau d\tau + V_N(t_0)$$

$$= \underbrace{\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N} \right)}_{C_{eq}} \int_{t_0}^t i \tau d\tau + V_1(t_0) + V_2(t_0) + \dots + V_N(t_0)$$

$$V = \frac{1}{C_{eq}} \int_{t_0}^t i \tau d\tau + V_{(t_0)}$$

The equivalent capacitance of N series capacitors is the reciprocal of the sum of the reciprocals of each capacitor.

Inductors

Passive element designed to store energy in its magnetic field.

Voltage across the inductor is proportional to the time rate of change of the current.

$$V = L \frac{di}{dt}$$

L : inductance of inductor Henry H

H: henry equal 1 volt-second per Ampere

$$L = \frac{N^2 \mu A}{l}$$

number of turns $\rightarrow N^2$

Permeability of the core $\rightarrow \mu$

Cross-section area $\rightarrow A$

length $\rightarrow l$

For linear inductors, the current-voltage relationship is:

$$V = L \frac{di}{dt} \Rightarrow di = \frac{1}{L} V dt$$

$$i = \frac{1}{L} \int_{-\infty}^t V \tau d\tau = \frac{1}{L} \int_{t_0}^t V \tau d\tau + i(t_0)$$

The power delivered to the inductor is:

$$P = v i = \left(L \frac{di}{dt} \right) i$$

The energy stored is:

$$\begin{aligned} W &= \int_{-\infty}^t P(\tau) d\tau = L \int_{-\infty}^t \frac{di}{d\tau} i d\tau \\ &= \int_{-\infty}^t di i = \frac{1}{2} L i^2(t) - \frac{1}{2} L i^2(-\infty) \end{aligned}$$

Since $i(-\infty) = 0$,

$$W = \frac{1}{2} L i^2$$

- Voltage across an inductor is zero when the current is constant.

-

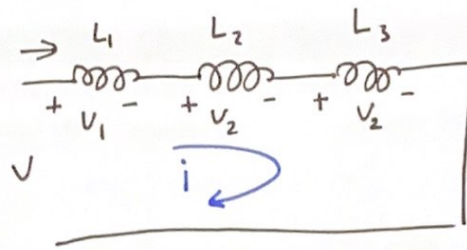
Series and Parallel Inductors

Using KVL

$$V = V_1 + V_2 + V_3$$

$$V = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt}$$

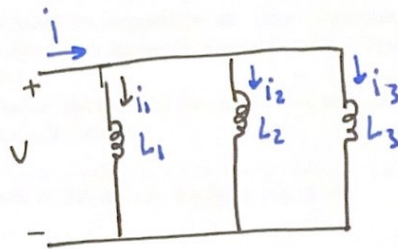
$$V = (L_1 + L_2 + L_3) \frac{di}{dt} = L_T \frac{di}{dt}$$



The equivalent inductance of series-connected inductors is the sum of the individual inductances.

Using KCL

$$i = i_1 + i_2 + i_3$$



$$i = \frac{1}{L_1} \int_{t_0}^t V dt + i_1(t_0) + \frac{1}{L_2} \int_{t_0}^t V dt + i_2(t_0) + \frac{1}{L_3} \int_{t_0}^t V dt + i_3(t_0)$$

$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right) \int_{t_0}^t V dt + i_1(t_0) + i_2(t_0) + i_3(t_0)$$

$$\Downarrow$$

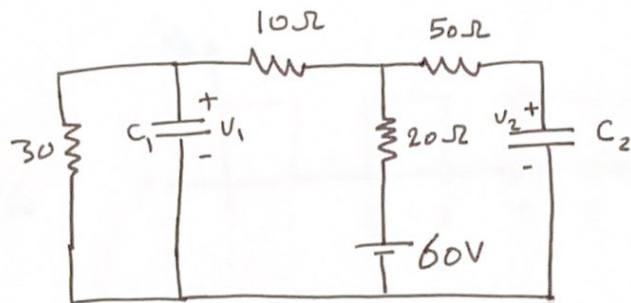
$$\frac{1}{L_{eq}}$$

The equivalent inductance of parallel-connected inductors is the reciprocal of the sum of the reciprocals of the individual inductances.

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} \quad \text{or} \quad L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

6.13 Find the voltage across the capacitors in the circuit under DC conditions.

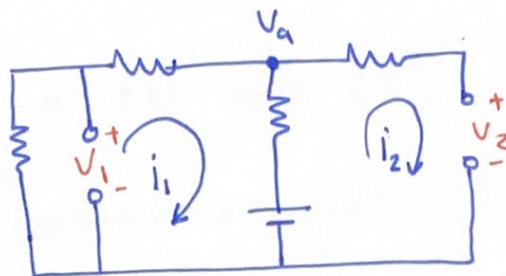
Under DC conditions
Capacitors react as open-
circuits.



$$i_2 = \underline{\underline{0}}$$

$$i_1 = \frac{V}{R_T} = \frac{60}{30+10+20} = \underline{\underline{1A}}$$

$$V_1 = V_{30\Omega} = \frac{30 * 60}{30+10+20} = \underline{\underline{30V}}$$



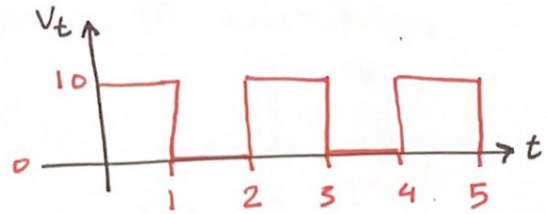
For V_2 , use KVL on V_a

$$\frac{V_a}{40} + \frac{V_a - 60}{20} = 0 \Rightarrow \frac{V_a + 2V_a - 120}{40} = 0$$

$$3V_a = 120 \Rightarrow \underline{\underline{V_a = 40V = V_2}}$$

6.42: Calculate the current through ~~the~~ ^{a 5H} inductor if the applied voltage is as the graph below.

Assume $i(0) = -1$ A



$$i(t) = \frac{1}{L} \int_{t_0}^t V_t dx + i_{t_0}$$

① $0 < t < 1$

$$i_t = \frac{1}{5} \int_0^t 10 dx + i(0) = 2 [x]_0^t - 1$$

$$i_t = 2t - 1 \quad [i(0) = -1, \quad i_1 = 2 \cdot 1 - 1 = 1]$$

② $1 < t < 2$

$$i_t = \frac{1}{5} \int_1^t 0 dx + i_1 = 1 \text{ A} \quad [i_1 = 1 \text{ A}, \quad i_2 = 1 \text{ A}]$$

③ $2 < t < 3$

$$i_t = \frac{1}{5} \int_2^t 10 dx + i(2) = 2 [x]_2^t + 1 = 2t - 4 + 1$$

$$= 2t - 3 \quad [i_2 = 2 \cdot 2 - 3 = 1 \text{ A}, \quad i_3 = 2 \cdot 3 - 3 = 3 \text{ A}]$$

$3 < t < 4$

$$i_t = \frac{1}{5} \int_3^t 0 dx + i_3 = 3 \text{ A} \quad [i_3 = 3 \text{ A}, \quad i_4 = 3 \text{ A}]$$

$4 < t < 5$

$$i_t = \frac{1}{5} \int_4^t 10 dx + i_4 = 2 [x]_4^t + 3 = 2t - 8 + 3$$

$$= 2t - 5$$

6.47: calculate the value of R that will make the energy stored in the capacitor the same as that stored in the inductor under DC conditions.

$$W_C = W_L$$

$$i_L = i_R = \frac{2 \times 5}{R+2} = \frac{10}{R+2}$$

$$V_C = V_R = i_R \cdot R = \frac{10R}{R+2}$$

$$W_C = \frac{1}{2} C V_C^2 = \frac{1}{2} \cdot 160 \times 10^{-6} \times \left(\frac{10R}{R+2} \right)^2$$

$$W_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times 4 \times 10^{-3} \times \left(\frac{10}{R+2} \right)^2$$

for $W_C = W_L$

$$\cancel{40} \times 10^{-6} \times \frac{\cancel{100} R^2}{(R+2)^2} = \cancel{2} \times 10^{-3} \times \left(\frac{\cancel{100}}{(R+2)^2} \right)$$

$$40 \times 10^{-3} R^2 = 1 \Rightarrow R^2 = \frac{1}{40 \times 10^{-3}} = \frac{1000}{40} = 25$$

$$R = 5 \Omega$$

